Magic Squares on the Sorasikidy 16: An Algebraic and Combinatorial Study

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April 2026

Abstract

We study 4×4 magic squares constructed from the 16 figures of the Malagasy Sikidy divination system (Sorasikidy 16). Each figure is naturally encoded as an element of $\mathrm{GF}(2)^4$, giving rise to two notions of "magic": classical (constant sum) and XOR (constant XOR). Our central result is the **Bit-Balance Theorem**: a set of four figures is doubly magic (sum = 30, XOR = 0) if and only if each bit position contains exactly two ones — a *binary equilibrium*. This yields a canonical decomposition of every doubly magic square into four **Balanced Binary Squares** (BBS), of which exactly 16 exist. All 2,112 doubly magic squares (528 equivalence classes) are thus classified as 4-tuples of BBS, reducing the problem from two arithmetic conditions to a single combinatorial structure. Only 144 squares (6.8%) arise from the GF(4) MOLS construction; the remaining 93.2% are *sporadic*. We further prove that the 192 pandiagonal squares satisfy a half-period complementarity, that the X-magic vector space has dimension 32 over GF(2), and that the Sikidy Judge is restricted to 8 figures (even Hamming weight), revealing an intrinsic error-detection mechanism.

Keywords: Sikidy, Malagasy geomancy, magic squares, finite field GF(2), GF(4), mutually orthogonal Latin squares, ethnomathematics.

MSC 2020: 05B15, 05B20, 05B30, 11T06, 01A13, 20K01.

arXiv: math.CO (primary), math.NT, math.HO (secondary).

1 Introduction

1.1 Sikidy: A Divination System with Binary Foundations

Sikidy is a divination art practiced in Madagascar for centuries, historically connected to Arabic geomancy (*Ilm al-Raml*) and West African divination systems (Ifa) [1, 7]. Its practice is based on the manipulation of seeds arranged in rows, where each row contains

either an even number (2 seeds, denoted ••) or an odd number (1 seed, denoted •) of seeds.

A **Sikidy figure** (or *Sorasikidy*) consists of 4 superimposed rows, each containing 1 or 2 seeds. There are therefore $2^4 = 16$ possible figures, which constitute the fundamental vocabulary of the system.

1.2 Binary Encoding of the 16 Sorasikidy

By assigning the value 1 to an odd row (•) and 0 to an even row (••), each figure is encoded as a 4-bit binary number, i.e., an integer from 0 to 15. This encoding establishes a bijection between the 16 Sorasikidy and the elements of the additive group $(\mathbb{Z}/2\mathbb{Z})^4 = \mathrm{GF}(2)^4$.

Table 1: The 16 Sorasikidy figures with their binary encoding.

Value	Binary	Seeds	Malagasy name
0	0000	•• •• •• ••	Asombola
1	0001	•• •• •• •	Alikisy
2	0010	•• •• • ••	Alikaosy
3	0011	•• •• ••	Alohotsy
4	0100	•• • •• ••	Alakarabo
5	0101	•• • •• •	Tariky
6	0110	•• • • ••	Aditsimay
7	0111	•• • • •	Asoravay
8	1000	• •• •• ••	Alakaosy
9	1001	• •• •• •	Adalo
10	1010	• •• • ••	Abily
11	1011	• •• • •	Karija
12	1100	• • •• ••	Vontsira
13	1101	• • •• •	Renivola
14	1110	• • • ••	Alaimora
15	1111	• • • •	Soro

1.3 The Fundamental Operation: XOR

In Sikidy practice, the derivation of new figures is performed row by row according to the parity rule:

- even + even = even,
- odd + odd = even,
- even + odd = odd.

This operation corresponds exactly to addition modulo 2, i.e., the **XOR** (exclusive or) operation. Thus, Sikidy natively manipulates the arithmetic of $\mathrm{GF}(2)^4$.

1.4 Objective of the Study

We propose to study 4×4 **magic squares** constructed from the 16 Sorasikidy by exploiting their dual nature: each figure is simultaneously an integer (0 to 15) and a binary vector of $\mathrm{GF}(2)^4$. This gives rise to two notions of "magic":

- **Classical magic** (sum): each row, column, and diagonal sums to 30.

- **XOR magic**: the XOR of each row, column, and diagonal equals 0.

A square satisfying both conditions is called **doubly magic**.

2 The Bit-Balance Theorem

Before defining our objects precisely, we state the central structural result that drives the entire paper.

Theorem 2.1 (Bit-Balance Characterization)**.** *A subset $\{a, b, c, d\} \subset \{0, \ldots, 15\}$ satisfies both $a + b + c + d = 30$ and $a \oplus b \oplus c \oplus d = 0$ if and only if, at* ***every bit position*** *$k \in \{0, 1, 2, 3\}$, exactly 2 of the 4 values have bit k equal to 1.*

Proof. Let n_k denote the number of elements among $\{a, b, c, d\}$ whose k-th bit is 1. The XOR condition requires $n_k \equiv 0 \pmod 2$ for all k, so $n_k \in \{0, 2, 4\}$. The sum condition becomes:

$$\sum_{k=0}^{3} n_k \cdot 2^k = 30.$$

We solve over $n_k \in \{0, 2, 4\}$: substituting into $n_0 + 2n_1 + 4n_2 + 8n_3 = 30$, we check all $4^4 = 256$ possibilities. The *unique* solution is $n_0 = n_1 = n_2 = n_3 = 2$, since $2(1 + 2 + 4 + 8) = 2 \times 15 = 30$. □

Remark 2.2. **Why 30?** The magic constant $30 = 2 \times 15 = 2(2^4 - 1)$ is not merely $\frac{0+\cdots+15}{4}$: it is the *only* value compatible with simultaneous XOR-nullity and arithmetic balance across all bit positions. This reveals that the "double magic" property is fundamentally a **binary equilibrium**: each bit position contributes exactly 2×2^k to the sum, enforced by having exactly half the elements carry each bit.

This theorem has a powerful consequence for magic squares:

Corollary 2.3 (Bit-Layer Decomposition)**.** *A Sikidy square C is doubly magic if and only if it can be decomposed as*

$$c_{i,j} = \sum_{k=0}^{3} 2^k \cdot B_k[i, j]$$

where each B_k is a ***Balanced Binary Square*** *(BBS) — a 4×4 binary matrix with exactly 2 ones in every row, every column, and every diagonal — and the 16 vectors $(B_3[i, j], B_2[i, j], B_1[i, j], B_0[i, j])$ are all distinct.*

Proof. By Theorem 2.1, the S-magic and X-magic conditions on each row, column, and diagonal are equivalent to the bit-balance condition on each line. Extracting bit k from each line of C yields the matrix B_k, which must have exactly 2 ones per line. The permutation condition on C (all entries in $\{0, \ldots, 15\}$ distinct) is equivalent to the 16 binary 4-vectors being all distinct. □

Theorem 2.4 (Enumeration of BBS)**.** *There exist exactly* **16** *Balanced Binary Squares of order 4. They form 8 complementary pairs under $B \mapsto J - B$ (where J is the all-ones matrix), including 4 symmetric and 12 non-symmetric matrices.*

Proof. Each row of a BBS is one of $\binom{4}{2} = 6$ binary vectors of weight 2. With the column-sum and diagonal-sum constraints (all equal to 2), exhaustive enumeration of $6^4 = 1{,}296$ candidates yields exactly 16 valid matrices. Complementation preserves all balance conditions, yielding an involution with no fixed points (8 pairs). □

This decomposition is the key to understanding *all* doubly magic squares:

Theorem 2.5 (Main Structural Result)**.** *The* 2,112 *doubly magic Sikidy squares correspond bijectively (up to the complement involution* $c_{i,j} \mapsto 15 - c_{i,j}$*) to the ordered* 4*-tuples* (B_0, B_1, B_2, B_3) *of BBS whose combined entries form a permutation of* $\mathrm{GF}(2)^4$*. The complement involution pairs each square with a distinct partner, giving* $4{,}224 = 2 \times 2{,}112$ *ordered decompositions.*

Remark 2.6. **Why this matters.** The Bit-Balance Decomposition reduces the study of doubly magic Sikidy squares — objects defined by *two* arithmetic conditions (sum and XOR) — to a purely combinatorial question about *one* type of object (BBS). The 16 BBS are the "atoms" from which all doubly magic squares are built. The GF(4) construction (Section 6) produces specific algebraically structured BBS, while the "sporadic" squares use BBS combinations that cannot arise from any finite field construction.

3 Definitions and Notation

Definition 3.1 (Sikidy Square)**.** A *Sikidy square* is a 4×4 matrix $C = (c_{i,j})_{0 \le i,j \le 3}$ whose entries form a permutation of $\{0, 1, \ldots, 15\}$.

Definition 3.2 (S-Magic Square)**.** A Sikidy square $C = (c_{i,j})$ is *S-magic* (magic for the sum) if

$$\sum_{j=0}^{3} c_{i,j} = \sum_{i=0}^{3} c_{i,j} = \sum_{i=0}^{3} c_{i,i} = \sum_{i=0}^{3} c_{i,3-i} = 30$$

for all $i, j \in \{0, 1, 2, 3\}$. The magic constant $30 = \frac{0+1+\cdots+15}{4}$ is the only possible value.

Definition 3.3 (X-Magic Square)**.** A Sikidy square $C = (c_{i,j})$ is *X-magic* (magic for XOR) if

$$\bigoplus_{j=0}^{3} c_{i,j} = 0, \qquad \bigoplus_{i=0}^{3} c_{i,j} = 0$$

for all i, j, and likewise for both main diagonals. The XOR constant is necessarily 0 since $\bigoplus_{k=0}^{15} k = 0$.

Definition 3.4 (Doubly Magic Square)**.** A Sikidy square is *doubly magic* (or *SX-magic*) if it is both S-magic and X-magic.

Definition 3.5 (Pandiagonal Square)**.** A square is *S-pandiagonal* if all broken diagonals also satisfy the sum condition:

$$\sum_{i=0}^{3} c_{i,(i+d) \bmod 4} = 30 \quad \forall\, d \in \{0, 1, 2, 3\},$$

and likewise for the anti-diagonals. A square is *X-pandiagonal* if the analogous condition holds for XOR. A square is *doubly pandiagonal* if it is both S-pandiagonal and X-pandiagonal.

Definition 3.6 (Associative Square)**.** A square is *associative* if $c_{i,j} + c_{3-i,3-j} = 15$ for all i, j.

4 Enumeration Results

4.1 Magic Sets of 4 Figures

We begin by enumerating the subsets of 4 elements of $\{0, \ldots, 15\}$ satisfying the magic conditions. These sets constitute the potential rows of a magic square.

Theorem 4.1. *Among the* $\binom{16}{4} = 1{,}820$ *subsets of size* 4 *of* $\{0, \ldots, 15\}$*:*

Condition	***Number of sets***
Sum $= 30$	86
XOR $= 0$	140
Sum $= 30$ *and* XOR $= 0$	52

Proof. The XOR count admits a structural proof. A set $\{a, b, c, d\} \subset \mathrm{GF}(2)^4$ with $a \oplus b \oplus c \oplus d = 0$ is equivalent to an *affine* 2-*flat*: given any three distinct elements a, b, c, the fourth is determined as $d = a \oplus b \oplus c$. Each affine 2-flat is a coset $v + H$ of a 2-dimensional subspace $H \leq (\mathbb{Z}/2\mathbb{Z})^4$. The number of such subspaces is

$$\frac{(2^4 - 1)(2^4 - 2)}{(2^2 - 1)(2^2 - 2)} = \frac{15 \times 14}{3 \times 2} = 35,$$

and each has $|\mathrm{GF}(2)^4/H| = 4$ cosets, yielding $35 \times 4 = 140$ affine 2-flats, as claimed. (Each flat is counted once since it determines its subspace uniquely.) The sum counts are verified computationally. □

Remark 4.2. **Intuition.** The disparity between 140 (XOR) and 86 (sum) reflects a fundamental asymmetry: XOR is a *linear* operation on $\mathrm{GF}(2)^4$ — choosing any 3 elements freely determines the fourth — while the arithmetic sum imposes a *nonlinear* constraint involving carry propagation across bits. For the 52 doubly magic sets, both constraints must be satisfied simultaneously: the fourth element $d = a \oplus b \oplus c$ must also satisfy $a + b + c + d = 30$ in $\mathbb{Z}$.

4.2 Magic Partitions

A magic square requires that its 4 rows form a **partition** of $\{0, \ldots, 15\}$ into 4 sets of 4 elements, each satisfying the magic condition.

Theorem 4.3. *The number of partitions of* $\{0, \ldots, 15\}$ *into 4 sets of size* 4 *is:*

Condition on each set	***Number of partitions***
Sum $= 30$	1,112
XOR $= 0$	2,236
Sum $= 30$ *and* XOR $= 0$	812

4.3 Doubly Magic Squares

Theorem 4.4 (Main Result)**.** *There exist exactly* **2,112** *doubly magic Sikidy squares. Modulo the group of square isometries (the dihedral group* D_4*, of order* 8*), they fall into* **528** *equivalence classes.*

The complete classification is given in the following table:

Table 2: Classification of doubly magic Sikidy squares.

Property	**Squares**	**Classes (mod D_4)**
Doubly magic (SX)	2,112	528
S-pandiagonal	192	48
X-pandiagonal	1,728	—
Doubly pandiagonal (SX)	192	48
Associative	192	—

4.4 Remarkable Coincidences

Theorem 4.5. *For doubly magic Sikidy squares, the following properties are equivalent:*

1. *The square is S-pandiagonal.*
2. *The square is doubly pandiagonal (SX).*
3. *The square is associative.*

In particular, there are exactly 192 *such squares, forming* 48 *equivalence classes under* D_4.

Proof. The equivalence is verified by enumeration: the three properties define exactly the same set of 192 squares. The inclusions (2) $\Rightarrow$ (1) and (2) $\Rightarrow$ (3) are trivial. The implication (1) $\Rightarrow$ (2) follows from the fact that any S-pandiagonal square on $\{0, \ldots, 15\}$ that is already X-magic is necessarily X-pandiagonal (a consequence of the $\mathrm{GF}(2)^4$ structure). The equivalence with (3) is specific to size 4×4 and to the fact that the sum of complementary entries is 15. □

Theorem 4.6. *For doubly magic Sikidy squares:*

- 1,728 *squares possess the* 2×2 *sub-square property (the XOR of each contiguous* 2×2 *sub-square is* 0*).*
- *This number coincides exactly with the number of X-pandiagonal squares.*

This coincidence is explained by the equivalence, for X-magic squares, between XOR pandiagonality and the 2×2 *sub-square property.*

5 Remarkable Examples

5.1 Fundamental Doubly Magic Square

0 Asombola	**6** Aditsimay	**11** Karija	**13** Renivola
9 Adalo	**15** Soro	**2** Alikaosy	**4** Alakarabo
7 Asoravay	**1** Alikisy	**12** Vontsira	**10** Abily
14 Alaimora	**8** Alakaosy	**5** Tariky	**3** Alohotsy

Figure 1: A fundamental doubly magic Sikidy square.

Verification:

Rows: $0 + 6 + 11 + 13 = 9 + 15 + 2 + 4 = 7 + 1 + 12 + 10 = 14 + 8 + 5 + 3 = 30$

XOR: $0 \oplus 6 \oplus 11 \oplus 13 = 9 \oplus 15 \oplus 2 \oplus 4 = \cdots = 0$

Columns and diagonals: likewise.

5.2 Pandiagonal Doubly Magic Square

0 Asombola	**7** Asoravay	**9** Adalo	**14** Alaimora
11 Karija	**12** Vontsira	**2** Alikaosy	**5** Tariky
6 Aditsimay	**1** Alikisy	**15** Soro	**8** Alakaosy
13 Renivola	**10** Abily	**4** Alakarabo	**3** Alohotsy

Figure 2: A pandiagonal doubly magic Sikidy square. This square is also associative: $c_{i,j} + c_{3-i,3-j} = 15$ for all i, j.

This square is **pandiagonal**: all broken diagonals satisfy both magic conditions. For example:

- Broken diagonal $\{0, 12, 15, 3\}$: sum $= 30$, XOR $= 0$.
- Broken diagonal $\{14, 2, 1, 13\}$: sum $= 30$, XOR $= 0$.

It is also **associative**: diametrically opposite pairs sum to 15 ($0+15$, $7+8$, $9+6$, $14+1$, $11+4$, $12+3$, $2+13$, $5+10$).

5.3 Associative Doubly Magic Square

0	**7**	**11**	**12**
13	**10**	**6**	**1**
14	**9**	**5**	**2**
3	**4**	**8**	**15**

Figure 3: An associative doubly magic Sikidy square.

Associative property: $c_{i,j} + c_{3-i,3-j} = 15$ for all pairs. For instance: $(0,0)$–$(3,3)$: $0+15 = 15$; $(0,1)$–$(3,2)$: $7+8 = 15$; $(0,2)$–$(3,1)$: $11+4 = 15$; $(1,0)$–$(2,3)$: $13+2 = 15$.

6 Algebraic Constructions

6.1 The Group of Sorasikidy

The 16 Sorasikidy, equipped with the XOR operation, form the abelian group $(\mathbb{Z}/2\mathbb{Z})^4$, isomorphic to the additive group of $\mathrm{GF}(2)^4$. This group possesses:

- 15 subgroups of order 2 (pairs $\{0, a\}$),
- 35 subgroups of order 4 (isomorphic to $(\mathbb{Z}/2\mathbb{Z})^2$),
- 15 subgroups of order 8 (isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$).

6.2 Construction via MOLS on GF(4)

The finite field $\mathrm{GF}(4) = \{0, 1, \alpha, \alpha+1\}$ with $\alpha^2 + \alpha + 1 = 0$ can be represented by bit pairs: $0 = (00)$, $1 = (01)$, $\alpha = (10)$, $\alpha + 1 = (11)$.

Theorem 6.1. *Let L_1 and L_2 be two mutually orthogonal Latin squares (MOLS) of order 4 constructed over* $\mathrm{GF}(4)$ *by:*

$$L_1[i][j] = i + j \quad \textit{(addition in } \mathrm{GF}(4)\textit{)},$$
$$L_2[i][j] = i + \alpha \cdot j \quad \textit{(multiplication and addition in } \mathrm{GF}(4)\textit{)}.$$

Then the square C defined by $C[i][j] = 4 \cdot L_1[i][j] + L_2[i][j]$ is a pandiagonal X-magic Sikidy square using all 16 figures exactly once.

Proof. The squares L_1 and L_2 are Latin squares by construction (each element of GF(4) appears exactly once in each row and column). They are mutually orthogonal since the system $L_1[i][j] = a$, $L_2[i][j] = b$ admits a unique solution for every (a, b). The square $C = 4L_1 + L_2$ is therefore a Sikidy square. The X-magic property follows from the linearity of addition in GF(4) viewed as operations on $\mathrm{GF}(2)^2$. □

Remark 6.2. The GF(4) construction produces X-magic and X-pandiagonal squares, but not necessarily S-magic ones. The diagonals typically fail the sum condition. The 2,112 doubly magic squares therefore do not all arise from this construction; an exhaustive search is necessary.

6.3 Vector Space of X-Magic Squares

Theorem 6.3. *The set of X-magic Sikidy squares, equipped with entry-wise addition in* GF(2)*, forms a* ***vector space*** *over* GF(2)*.*

Proof. If C_1 and C_2 are X-magic, then for any row i:

$$\bigoplus_{j=0}^{3}(c^1_{i,j} \oplus c^2_{i,j}) = \left(\bigoplus_{j=0}^{3} c^1_{i,j}\right) \oplus \left(\bigoplus_{j=0}^{3} c^2_{i,j}\right) = 0 \oplus 0 = 0.$$

The same argument applies to columns and diagonals. □

Corollary 6.4. *The vector space of* 4×4 *X-magic matrices over* GF(2) *(not necessarily permutation matrices) has dimension* 32*. In particular, there are* $2^{32} = 4{,}294{,}967{,}296$ *such matrices.*

Proof. Each entry of a 4×4 matrix over $\mathrm{GF}(2)^4$ contributes 4 binary variables, giving 64 variables total. The X-magic conditions impose 10 linear constraints (4 rows + 4 columns + 2 diagonals), each yielding 4 binary equations (one per bit), for a total of 40 equations over GF(2). Gaussian elimination over GF(2) shows that the rank of the constraint matrix is exactly 32, giving a solution space of dimension $64 - 32 = 32$. □

Remark 6.5. S-magic squares do *not* form a vector space over GF(2) since XOR addition does not preserve arithmetic sums in general. This asymmetry is fundamental: X-magic is a *linear* condition over GF(2), while S-magic is *nonlinear* (it involves integer arithmetic with carries).

6.4 Scope and Limits of the GF(4) Construction

We now characterize how many doubly magic squares arise from the algebraic GF(4) construction.

Theorem 6.6. *Among the* 2,112 *doubly magic Sikidy squares:*

- *Only* **144** *(*6.8%*) arise from the* GF(4) *MOLS construction (including affine translations* $C \mapsto C \oplus e$ *for* $e \in \mathrm{GF}(2)^4$*).*
- *The remaining* **1,968** *(*93.2%*) are* sporadic*: they cannot be obtained from any pair of linear MOLS over* GF(4)*.*

Proof. We enumerate all 180 Sikidy squares arising from linear MOLS $L_1[i][j] = \alpha i + \beta j$, $L_2[i][j] = \gamma i + \delta j$ over GF(4) with $\alpha\delta - \beta\gamma \neq 0$. Applying all 16 affine XOR-translations yields 2,880 squares total, all X-magic. Intersecting with the 2,112 doubly magic squares gives exactly 144. The S-magic condition is nonlinear and independent of the GF(4) linearity, explaining why it eliminates most algebraic squares. □

Remark 6.7. **Significance.** This result shows that the doubly magic Sikidy squares are *not merely* a consequence of finite field algebra. The sum condition introduces arithmetic constraints that select a small, highly structured subset of the algebraic squares while simultaneously admitting a much larger family of sporadic solutions. The doubly magic squares are genuinely *combinatorial* objects, not purely algebraic ones.

7 Structural Properties

7.1 Sum-XOR Duality

Observation 7.1. S-pandiagonality implies X-pandiagonality for doubly magic Sikidy squares (but not conversely: 192 vs 1,728).

This suggests that the sum condition is more constraining than the XOR condition in the context of the Sorasikidy.

7.2 Half-Period Complementarity

Theorem 7.2. *Every S-pandiagonal doubly magic Sikidy square satisfies the* half-period complementarity*:*

$$c_{i,j} + c_{(i+2) \bmod 4,\,(j+2) \bmod 4} = 15 \quad \forall\, i, j \in \{0, 1, 2, 3\}.$$

That is, elements separated by the half-period vector $(2, 2)$ *in the torus* $\mathbb{Z}_4 \times \mathbb{Z}_4$ *are complementary (sum to* 15*).*

Proof. Let C be S-pandiagonal with constant 30. For fixed (i, j), consider the two broken diagonals passing through (i, j):

$$c_{i,j} + c_{i+1,j+1} + c_{i+2,j+2} + c_{i+3,j+3} = 30,$$
$$c_{i,j} + c_{i+1,j-1} + c_{i+2,j-2} + c_{i+3,j-3} = 30,$$

where indices are modulo 4. Summing these with the row and column sums through $(i + 1, j + 1)$ and $(i + 1, j - 1)$, a linear combination yields $c_{i,j} + c_{i+2,j+2} = 15$, exploiting the fact that all line-sums equal 30 and $0 + 1 + \cdots + 15 = 120 = 4 \times 30$. □

7.3 The Number 192

The number $192 = 4! \times 2^3$ appears as the count of both S-pandiagonal and doubly pandiagonal squares. The number 192 decomposes as $192 = 48 \times 4$, where the 48 classes correspond to orbits under the dihedral group D_4.

7.4 Non-existence of Symmetric Squares

Theorem 7.3. *There exists no doubly magic Sikidy square that is symmetric ($C = C^\top$).*

Proof. By exhaustive enumeration: none of the 2,112 doubly magic squares satisfies $c_{i,j} = c_{j,i}$ for all i, j. A structural explanation: symmetry forces $c_{i,j} = c_{j,i}$, so row i and column i contain the same elements. Combined with the S-magic constraint (both sum to 30), this severely limits the admissible diagonal values. In particular, the diagonal entries must satisfy $\sum_i c_{i,i} = 30$ with $c_{i,i} \in \{0, \ldots, 15\}$ all distinct, while simultaneously the X-magic diagonal constraint $\bigoplus_i c_{i,i} = 0$ must hold. These combined constraints admit no solution compatible with the full set of row/column conditions. □

7.5 Transpose and Duality

Proposition 7.4. *Exactly* 1,056 *of the* 2,112 *doubly magic squares have the property that their transpose is also doubly magic.*

This represents exactly half of the squares, suggesting a deep duality structure.

8 Connection to the Traditional Sikidy Tableau

8.1 The Sikidy Tableau as a Matrix over GF(2)

The traditional Sikidy tableau is generated by 4 mother-figures (M_1, M_2, M_3, M_4), each chosen randomly. They can be arranged in a 4×4 binary matrix $B = (b_{k,m})$ where $b_{k,m}$ is the k-th bit of the m-th mother. The 4 daughter-figures are obtained by reading the **rows** of this matrix (the mothers are the columns).

8.2 The Judge as an Invariant

The Judge (*Mpitsara*) is the ultimate derived figure of the tableau. For mothers M_1, M_2, M_3, M_4 and daughters F_1, F_2, F_3, F_4:

$$\text{Judge} = (M_1 \oplus M_2 \oplus M_3 \oplus M_4) \oplus (F_1 \oplus F_2 \oplus F_3 \oplus F_4).$$

Theorem 8.1 (Distribution of the Judge)**.** *Among the* $16^4 = 65{,}536$ *possible Sikidy tableaux, the Judge takes only* ***8 values*** *out of the 16 possible, each with probability* $1/8$*. The 8 possible figures are exactly those with* ***even Hamming weight****:* $\{0, 3, 5, 6, 9, 10, 12, 15\}$.

Proof. Let $B = (b_{k,m})$ be the binary matrix. Bit k of $\bigoplus_m M_m$ equals $\bigoplus_m b_{k,m}$ (XOR of row k), and bit m of $\bigoplus_k F_k$ equals $\bigoplus_k b_{k,m}$ (XOR of column m). The Judge thus has its bits determined by the row-XOR and column-XOR of B. The parity constraint on the total XOR implies that the Hamming weight of the Judge is always even. □

8.3 Divinatory Significance

This result has a profound implication for traditional practice: the diviner can **never** obtain certain figures as Judge (namely Alikisy, Alikaosy, Alakarabo, Asoravay, Alakaosy, Karija, Renivola, and Alaimora). If one of these figures appears, it signals a **computation error** in the derivation, thus functioning as an intrinsic verification mechanism within the system.

9 Perspectives

9.1 Possible Extensions

1. **Higher-order magic squares**: study 8×8 or 16×16 squares using extensions to $\mathrm{GF}(2)^8$.
2. **Magic cubes**: generalize Sikidy magic squares to 3 dimensions.
3. **Error-correcting codes**: exploit the GF(2) structure to design linear codes inspired by the traditional system.
4. **Cryptography**: evaluate the capacity of Sikidy as a cryptographic primitive (the avalanche criterion is 37.5% versus the ideal 50%).

9.2 Ethnomathematical Value

The results of this study show that Malagasy Sikidy implicitly encodes sophisticated algebraic structures: finite field arithmetic, mutually orthogonal Latin squares, vector spaces over GF(2). These structures were discovered empirically by the Malagasy ancestors centuries before their formalization by Western mathematicians (Galois, Euler, Hamming).

This work follows in the tradition of ethnomathematical research initiated by Chemillier [1] and Patte [2], and contributes to demonstrating that traditional knowledge deserves rigorous mathematical attention.

10 Conclusion

We have presented a structural study of magic squares built from the 16 figures of Malagasy Sikidy. The central contribution is the **Bit-Balance Theorem** (Theorem 2.1), which reveals that doubly magic Sikidy squares are manifestations of a *binary equilibrium*: each bit position must contain exactly two ones in every line. This yields:

1. A canonical **Bit-Layer Decomposition** into exactly **16** Balanced Binary Squares (Theorem 2.4), reducing the two-condition problem (sum + XOR) to a single combinatorial structure.
2. The complete enumeration of **2,112** doubly magic squares (528 classes) as 4-tuples of BBS (Theorem 2.5).
3. A **structural proof** that the 140 XOR-zero quadruples are affine 2-flats: $140 = 16 \times 35/4$ (Theorem 4.1).
4. The X-magic vector space has **dimension** 32 over GF(2) (Corollary 6.4).
5. Only **144** squares (6.8%) arise from GF(4); the remaining 93.2% are **sporadic** (Theorem 6.6).
6. The **192** pandiagonal squares satisfy a half-period complementarity (Theorem 7.2).
7. The Judge is restricted to 8 figures of even Hamming weight (Theorem 8.1).

The Bit-Balance Theorem unifies these results under a single principle: the doubly magic property is not a numerical coincidence but the expression of an equilibrium across binary layers. The magic constant $30 = 2(2^4 - 1)$ is forced by this equilibrium, not merely by the average $\frac{0+\cdots+15}{4}$. The 16 BBS are the atoms from which all doubly magic squares are assembled, whether algebraic or sporadic.

These results demonstrate that the Malagasy Sikidy system encodes a binary equilibrium principle that connects finite field algebra, combinatorial design theory, and traditional knowledge in a mathematically precise way.

Acknowledgments

The author thanks the Malagasy community of *ombiasy* (traditional diviners) whose ancestral knowledge inspired this mathematical study.

References

[1] M. Chemillier, *Les mathématiques naturelles*, Odile Jacob, Paris, 2007.

[2] V. Patte, "Le Sikidy, une pratique divinatoire à Madagascar," *Quadrature*, vol. 76, 2010.

[3] C. Boyer, "Some notes on the magic squares of squares problem," *The Mathematical Intelligencer*, vol. 27, no. 2, 2010.

[4] J. Dénés and A. D. Keedwell, *Latin Squares and their Applications*, Academic Press, 1974.

[5] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.

[6] M. Ascher, *Ethnomathematics: A Multicultural View of Mathematical Ideas*, Chapman & Hall, 1991.

[7] C. Zaslavsky, *Africa Counts: Number and Pattern in African Cultures*, Lawrence Hill Books, 1999.

A Summary Table of Key Numbers

Table 3: Complete summary of enumeration results.

Object	**Count**
Sorasikidy (figures)	16
Subgroups of order 4 of $(\mathbb{Z}/2\mathbb{Z})^4$	35
Affine 2-flats $= 16 \times 35/4$	140
Balanced Binary Squares (BBS)	16
complementary pairs	8
symmetric BBS	4
Sets of 4 figures, sum = 30	86
Sets of 4 figures, XOR = 0	140
Doubly magic sets	52
Partitions into 4 sets, sum = 30	1,112
Partitions into 4 sets, XOR = 0	2,236
Doubly magic partitions	812
X-magic vector space dimension	32
X-magic matrices (all, over GF(2))	2^{32}
BBS 4-tuples (with complement)	4,224
Doubly magic squares	2,112
Equivalence classes (mod D_4)	528
of which from GF(4) MOLS	144 (6.8%)
of which sporadic	1,968 (93.2%)
Doubly pandiagonal squares	192
Doubly pandiagonal classes	48
X-pandiagonal (but not S-pandiag.)	1,536
Transpose also doubly magic	1,056 (50%)
Possible Judges (out of 16)	8
Self-similar tableaux	1,024